Analyzing Crime Data with Applied Machine Learning Models

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Motivation/Relevance:

According to Wikipedia, New Mexico is rated second for the highest violent crime rate per 100k in the country [6]. We are motivated to explore the factors that influence crime and to identify areas for improvement in New Mexico. We are proposing a project that will use state and national crime data to identify features of crime and/or reveal interesting patterns. Our goal is to support New Mexico policymakers and law enforcement in combating crime more effectively by offering data-backed insights.

Problem:

To identify the factors that contribute to New Mexico's high crime rate, we must analyze state crime data and compare it to other states. While current research supports socioeconomic status influencing crime rate [3], we are also interested in revealing other factors that also influence crime. By doing so, we can identify areas of improvement for policymakers. In this project, we would like to focus on comparing crime data between states and use datasets from the most recent years available (2022-2024) on the FBI Crime Data Explorer. Our project will compare New Mexico with Colorado since it is a neighboring state and has a relatively lower crime rate [6]. We recognize the implications our work may have and will approach the topic sensitively and ethically.

Applied Machine Learning Tasks:

To identify the demographic factors that influence crime, we will employ a logistic regression model to classify which states have high or low crime rates using demographic factors such as education, income, age, sex, and employment. Our main interest in the logistic regression model is the weights per feature. From the weights, we can determine which feature, in this case, holds the most influence on the high crime rate.

To identify areas of high crime, we will employ K-Nearest Neighbors clustering using FBI crime location data. We can then identify areas in need of more police support. We can also use census data to map counties with low or high income or other demographic factors to visualize a relationship between areas of high or low crime and related demographic factors.

We would like to employ more machine learning models, and as we progress in the course, we draw more inspiration on which models to use. In particular, we would like to explore relationships between income brackets or educational attainment, and the type of crime committed. We have purposely excluded race and ethnicity from our analysis to avoid racial bias/implications.

Dataset Selection and Justification:

The FBI crime database collects data from all 50 states and a variety of state agencies that serve those state populations. We are starting our analysis with the arrest records from Colorado and New Mexico. The arrest record contains information about the arrest (when it took place and the type of offense) and the arrestee (age, sex, race, and ethnicity). We can use this data to analyze both violent and non-violent crime rates.

Separate from the arrestee data, there are also records quantifying the total number of arrests made by each agency in each state. This data can be pulled from the FBI crime database as a .json file, but each agency must be called individually. We wrote code that could loop through all the agencies for a state and compile the desired elements of the .json files into a working dataframe for our future analysis. The goal for this dataset is to support a clustering analysis and look at the

crimes by location. The total population served by a particular agency is stored along with the total arrests to support calculating the crime rate for a region and comparing it to national rates and other state rates.

The above datasets are suitable for our analysis because they are the most comprehensive collection of this type of data. We selected the most recent year (2024) for which data had been published. These datasets are complete (as much as possible). These datasets are well-designed and consistent. Once we had written a few lines of code to compile the .json files, it was repeatable across all 257 state agencies in Colorado and 135 in New Mexico. It is also worth mentioning that the populations supported by these agencies sum to a value that exactly matches the total population for each state listed in the census estimations for 2024.

In addition to the FBI crime datasets, we collected demographic data from the United States Census Bureau. We collected 4 different tables: Selected Economic Characteristics, Income in the past 12 months, ACS Demographic and Housing estimates, and Educational Attainment for all 50 states, New Mexico, and Colorado, respectively. To generate the tables using the census data webpage, we applied the following filter: year = 2024 and state = "All 50 States". For New Mexico and Colorado, the state is replaced with county = "All Counties". County-level data is incredibly limited. The census data does not have data on all counties within New Mexico and Colorado. Additionally, the datasets listed above are from the yearly "American Community Survey" since the Decennial Census does not provide demographic data such as education attainment, income, age, sex, and employment. This is also advantageous to our analysis because the most recent data available is from 2024, just like the FBI Crime data.

Using the Python language, we have loaded the collected datasets from the US Census Bureau data page into a Jupyter Notebook. The notebook titled "Phase3_CENSUS.ipynb" loads the datasets, reformats the datasets since they are output from the webpage as nested pivot tables, and creates cleaned (as well as possible) new datasets for income, employment, and age and sex metrics per state. A dataset for education is not created at this point, as the table output from the website is incredibly nested, and we needed more time to explore ways to flatten it.

The "Income in the past 12 months" data table has details regarding income brackets per household type per state(or county, depending on New Mexico or Colorado). It includes the total estimate, total percent, and total margin error for both estimate and percent. The dataset for all states in the United States has 17 rows and 417 columns. The dataset for New Mexico has 17 rows and 73 columns. The dataset for Colorado has 17 rows and 97 columns.

The "Educational Attainment" data table has details regarding educational attainment per demographic type per state(or county, depending on New Mexico or Colorado). This analysis will mainly use the demographic type of "age by educational attainment". It includes the total estimate, total percent, total percent male, total percent female, and total margin error for all metrics. The dataset for all states in the United States has 20 rows and 625 columns. The dataset for New Mexico has 20 rows and 109 columns. The dataset for Colorado has 20 rows and 145 columns.

The "Selected Economic Characteristics" data table has details regarding economic factors per state(or county, depending on

New Mexico or Colorado). This analysis will mainly use the demographic type of "age by educational attainment". It includes the total estimate, total percent, and total margin error for all metrics. The dataset for all states in the United States has 20 rows and 209 columns. The dataset for New Mexico has 20 rows and 37 columns. The dataset for Colorado has 20 rows and 49 columns.

The "ACS Demographic and Housing estimates" data table has details regarding demographic types per state(or county, depending on New Mexico or Colorado). This analysis will mainly use the demographic type of "age and sex". We have purposely excluded "race and ethnicity" to avoid potential ethical implications. It includes the total estimate, total percent, and total margin error for all metrics. The dataset for all states in the United States has 20 rows and 625 columns. The dataset for New Mexico has 20 rows and 37 columns. The dataset for Colorado has 20 rows and 49 columns.

Combined Datasets:

For Phase 4 of this project, we started working to combine the FBI crime datasets with our selected demographic data for the total US population. From the demographic data, we combined the predictive factors (Table 3) with "Total all classes, "Violent crime" of the FBI US Crime dataset:

After combining the data, we created two new columns regarding "Crime Rate" and "Violent Crime Rate". Of the combined dataset, there are 51 rows (the 50 states and the District of Columbia) and 56 columns. We then created three classes of low, medium, and high crime after calculating the crime rate per state and added a new column "Crime Rate Class" to the dataset. The rest of the crime dataset will be set aside for a separate analysis later. Our target features are "Crime Rate Classes" and "Violent Crime".

Methods:

Using the models available in scikit-learn we employed Linear Regression, Logistic Regression, Non-linear SVM, Decision Tree Classifier, and K-NearestNeighbors.

Each model was employed on the predictive features of

The target class for linear regression was violent crime while the rest was low, medium, or high crime rate. If applicable, all models were initialized with a random state of 1 to ensure reproducibility.

For linear regression, percentages of each feature had to be converted into total numbers. This was done by multiplying the percentage by the "Totals" features such as 'Total population', 'Total Households', or 'Population 16 years and over' depending on whether it was income, education, or employment features. This was not done for the other models.

The logistic regression model was trained using L2 (Ridge) regularization, with an inverse regularization strength $C=0.5C = 0.5$C=0.5 and a maximum of 200 iterations to ensure convergence.

We trained a non-linear support vector machine (SVM) using the radial basis function (RBF) kernel, with a gamma value of 0.10 to control the influence of individual data points, and a regularization parameter CCC of 10.0 to balance margin width and classification error.

The decision tree classifier was trained using the Gini impurity criterion to measure splits and a maximum tree depth of 10.

The K-Nearest Neighbors (KNN) model was trained using 7 neighbors and the Euclidean distance metric to measure similarity between data points.

In seperate runs, Principal Component Analysis (PCA) was applied to the linear and logistic regression models, using 10 components. This number of components was chosen as it explained the majority of the variance in the data (Figure 3). Kernel PCA (kPCA) was applied to the non-linear SVM model with 10 components, an RBF kernel, and gamma set to 0.25. No dimension reduction techniques were applied to the Decision Tree Classifier or K-Nearest Neighbors models.

Results and Observations:

Overall, the predictive performance of all models was limited. Linear regression achieved a moderate R^2 of 0.43 without PCA, but performance dropped with PCA. For classification tasks, logistic regression achieved only 27% accuracy, while non-linear SVM performed slightly better at 54% accuracy, and decision tree and K-Nearest Neighbors achieved 45–46% accuracy. Dimension reduction techniques (PCA and kPCA) did not substantially improve performance.

Model training and testing were generally fast, with linear regression and KNN completing in under 0.002 seconds, and dimension reduction for the non-linear SVM taking the longest at 0.012 seconds.

The results suggest that the current feature set is insufficient to reliably predict crime rate classes, likely due to skewed distributions (Figure 4) and a small dataset (51 rows of state level data).

Although the models achieved relatively low accuracy and R^2 scores, the linear and logistic regression models still offered valuable information regarding feature importance, highlighting which variables contributed most to predicting crime rate classes (Figure 5 and 6). 35 to 44 years olds who hold a high school diploma or higher are more likely to predict violent crime, coefficient of 176945.575966 while the unemployed population is less likely to predict violent crime -112207.585147 (Figure 5). As for states with high crime rates, heavily weighted coefficient features included a negative coefficient of -0.663496 for 85-year-olds or older and a positive coefficient of 0.434142 for 25- to 34-year-olds.

Issues to Address:

The current dataset is limited in size, making it difficult to perform a train-test split and obtain reliable insights from regression or classification analyses. Moreover, the large number of predictive features may need to be reduced, either through dimension reduction techniques or by grouping variables into meaningful brackets. The data is also heavily left-skewed, indicating that most states have moderate to low crime rates, which makes creating balanced classes a challenge. To improve predictive performance, additional data may be needed by expanding into county-level data across all 50 states, generating hypothetical state-level observations, or incorporating multiple years of census data. Reducing feature complexity and broadening the dataset could provide more robust and interpretable results in future analyses.

Related works:

Oatley [1] examines how AI and big data technologies are influencing major issues in crime analytics, such as predictive policing, legal decisions, bias in profiling, crime forecasting, and AI regulation.

Onyenke & Karam [2] examine how economic deprivation and residential mobility influence crime in vulnerable urban neighborhoods. Through semi-structured interviews, it finds that socioeconomic inequality contributes to rising delinquency.

Roshankar & Keyvanour [3] introduce the Geographic Crime Analysis Framework (G.C.A.F.), a spatial–temporal approach designed to improve the analysis and prediction of urban crime. The framework compares different prediction methods, evaluates their strengths and weaknesses, and provides criteria to help law enforcement and policymakers choose the most effective approach.

Wikipedia contributors [4] list the states by violent crime rate per 100k.

Figures:

Top 10 New Mexico Arrest Offenses	
Simple Assault	10801
Drug/Narcotic Violations	4828
Shoplifting	3696
Aggravated Assault	2945
Destruction/Damage/Vandalism of Property	1532
Stolen Property Offenses	1014
Drug Equipment Violations	888
Burglary/Breaking & Entering	837
Weapon Law Violations	779
All Other Offenses	440

Table 1: Arrest Offenses for New Mexico

Top 10 Colorado Arrest Offenses	
Simple Assault	15080
Drug/Narcotic Violations	12436
Shoplifting	11555
Aggravated Assault	7303
Drug Equipment Violations	6373
Destruction/Damage/Vandalism of Property	4911
All Other Offenses	4721
All Other Larceny	4157
Motor Vehicle Theft	3060
Burglary/Breaking & Entering	2950

Table 2: Arrest Offenses for Colorado

Totals	'Total population' ‘Total Households’ ‘Population 16 years and over’
Age	'Under 5 years', '5 to 9 years', '10 to 14 years', '15 to 19 years', '20 to 24 years', '25 to 34 years', '35 to 44 years', '45 to 54 years', '55 to 59 years', '60 to 64 years', '65 to 74 years', '75 to 84 years', '85 years and over'
Sex at Birth	'Female', 'Male'
Employment	'In labor force', 'Not in labor force'
Education	‘Less than HS graduate’ ‘HS graduate (includes equivalency)’ ‘Some college’ ‘Bachelor’s degree or higher’ Each of these brackets is broken into a sub-bracket by age: ‘18 to 24 years’ '25 to 34 years’ ‘35 to 44 years’ ‘45 to 64 years’ ‘65 years and over’ Alternate sub-brackets: ‘18 to 24 years’

	‘25 years and over’
Income	'Less than $10,000', '$10,000 to $14,999', '$15,000 to $24,999', '$25,000 to $34,999', '$35,000 to $49,999', '$50,000 to $74,999', '$75,000 to $99,999', '$100,000 to $149,999', '$150,000 to $199,999', '$200,000 or more'

Table 3: Predictive Factors from US Census Data 2024

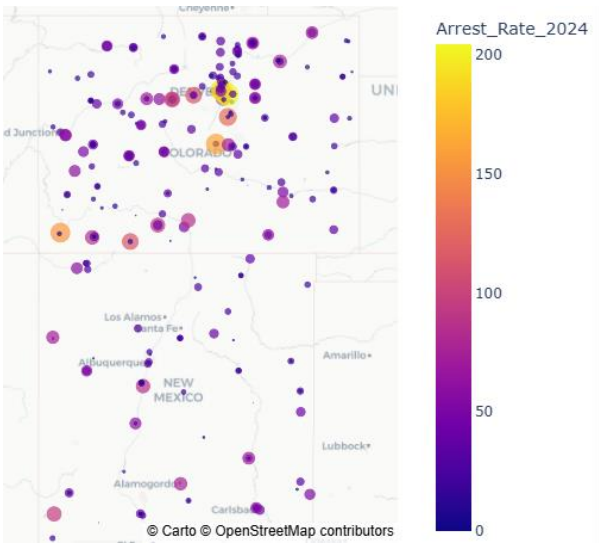

Figure 1: Arrest Rate by agency in New Mexico and Colorado

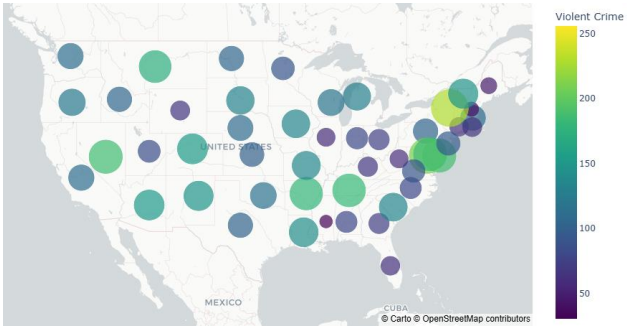

Figure 2: Violent Crime Rate by state for the US

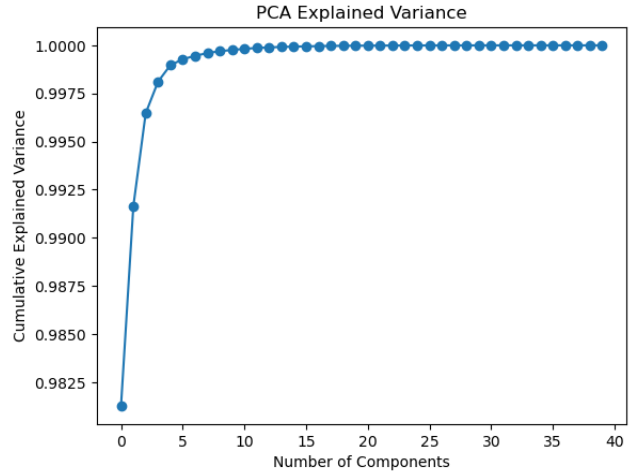

Figure 3: PCA Explained Variance

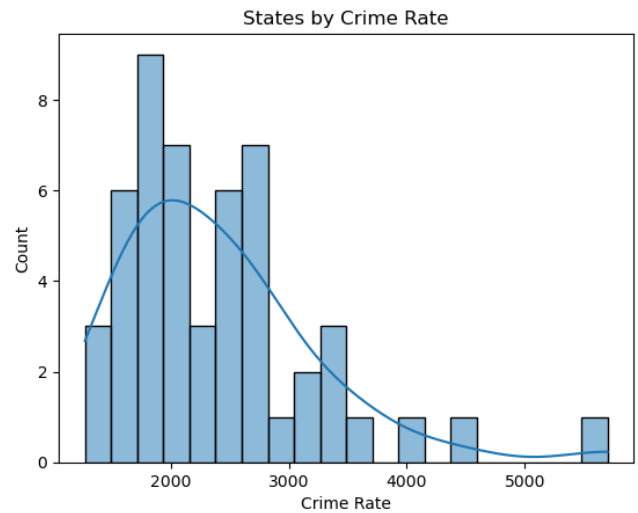

Figure 4: Histogram of States by Crime Rate

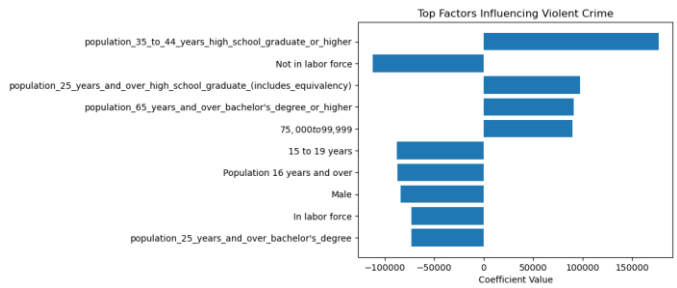

Figure 5: Top Factors Influencing Violent Crime

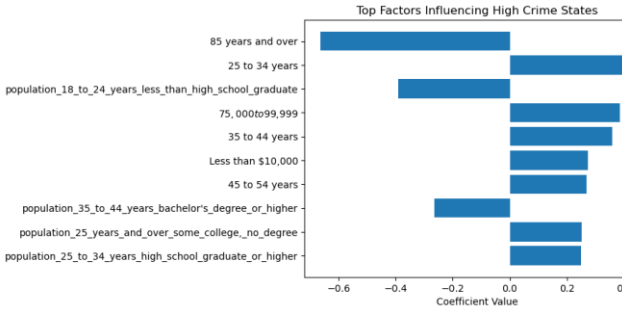

Figure 6: Top Factors Influencing Violent Crime

Sources:

1. Oatley, G.C. 2021. *Themes in data mining, big data, and crime analytics.* Retrieved from: https://wires.onlinelibrary.wiley.com/doi/abs/10.1002/widm.1432 (Accessed: 05 February 2026).

2. Onyenke, C.C. & Karam, A.H. 2022. *An Exploratory Study of Crime: Examining Lived Experiences of Crime through Socioeconomic, Demographic, and Physical Characteristics. Urban Sci, 6(3), 43.* Retrieved from: https://doi.org/10.3390Retrievedi6030043 (Accessed: 05 February 2026)

3. Roshankar, R., and Keyvanpour, M.R. 2025. *GeoCrime Analytic Framework (G.C.A.F.): A Comprehensive Framework for Dynamic Spatial Temporal Crime Analysis.* Retrieved from: https://link.springer.com/article/10.1007/s12061-025-09640-y (Accessed: 05 February 2026).

4. Wikipedia contributors. 2025. List of U.S. states and territories by violent crime rate. In

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5.FBI Crime Data Explorer

   https://cde.ucr.cjis.gov/LATEST/webapp/#/pages/explorer/crime/crime-trend

6. City of Albuquerque Maps https://www.cabq.gov/gis/

7. Census Data

   https://data.census.gov/table